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(54) **SOLAR CELL**

(71) Applicant: **Trina Solar Co., Ltd.**, Changzhou  
(CN)

(72) Inventors: **Changseo Park**, Seoul (KR); **Yoonsil Jin**, Seoul (KR); **Youngho Choe**, Seoul (KR)

(73) Assignee: **Trina Solar Co., Ltd.**, Changzhou  
(CN)

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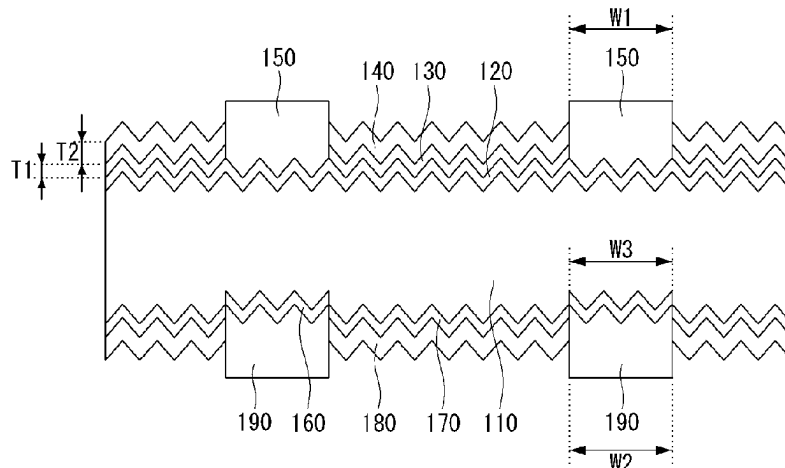
*Primary Examiner* — Sadie White

(74) *Attorney, Agent, or Firm* — Taft Stettinius & Hollister LLP

(57) **ABSTRACT**

A bifacial solar cell includes a silicon substrate; an emitter layer; a plurality of first electrodes locally on the emitter layer; a first aluminum oxide layer on the emitter layer; a first silicon oxide layer between the first aluminum oxide layer and the emitter layer; a first anti-reflection layer on the first aluminum oxide layer; a back surface field layer on the silicon substrate; a second aluminum oxide layer on the silicon substrate; a second silicon oxide layer between the second aluminum oxide layer and the silicon substrate; a second anti-reflection layer on the second aluminum oxide layer; and a plurality of second electrodes respectively on the back surface field layers through the second anti-reflection layer, the second aluminum oxide layer and the second silicon oxide layer.

**19 Claims, 4 Drawing Sheets**



**Related U.S. Application Data**

continuation of application No. 15/443,331, filed on Feb. 27, 2017, now Pat. No. 10,446,697, which is a continuation of application No. 13/182,203, filed on Jul. 13, 2011, now Pat. No. 9,608,133.

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FIG. 1

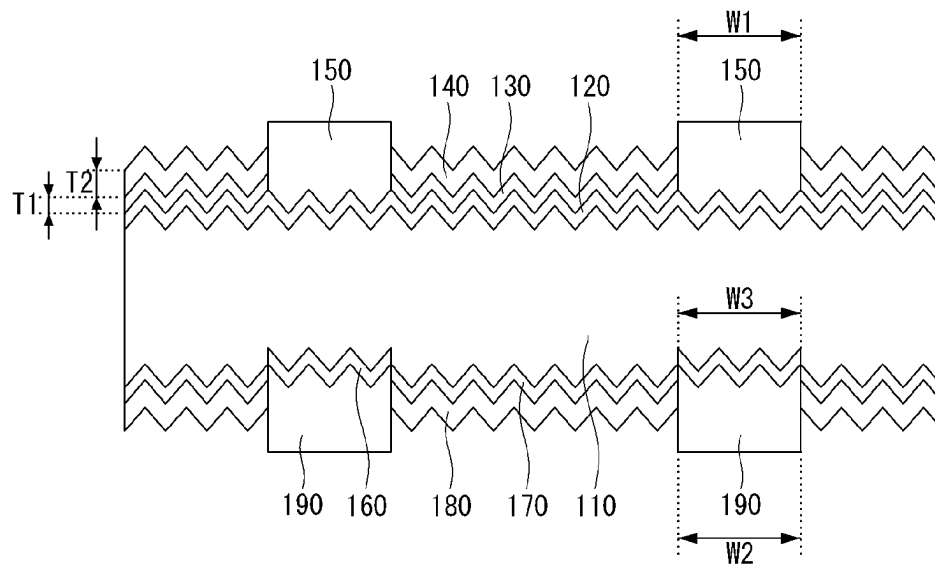
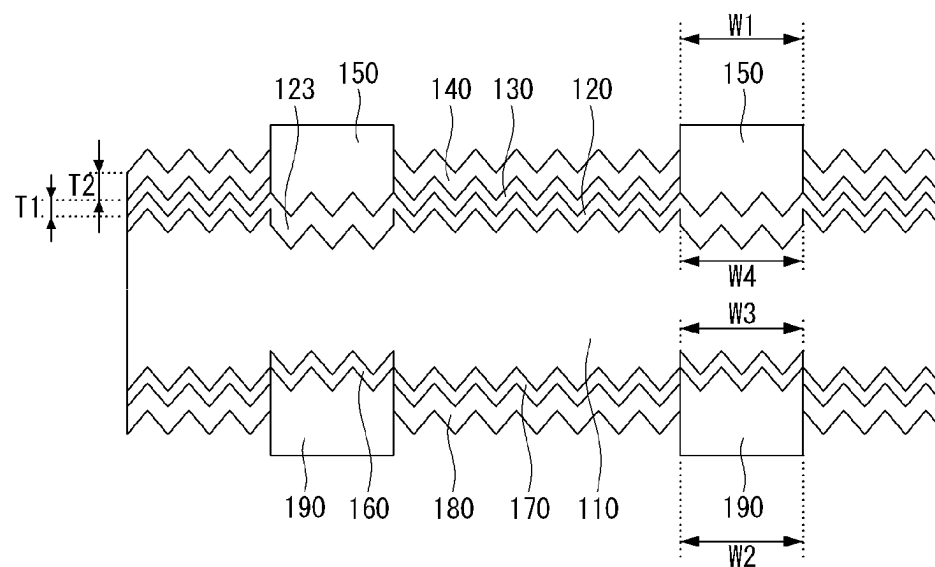
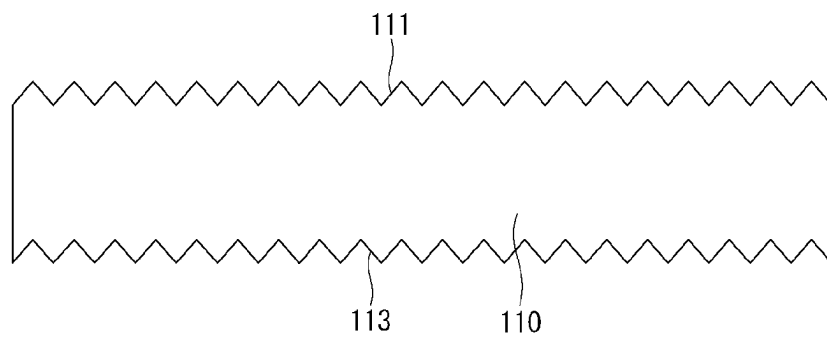


FIG. 2



**FIG. 3**



**FIG. 4**

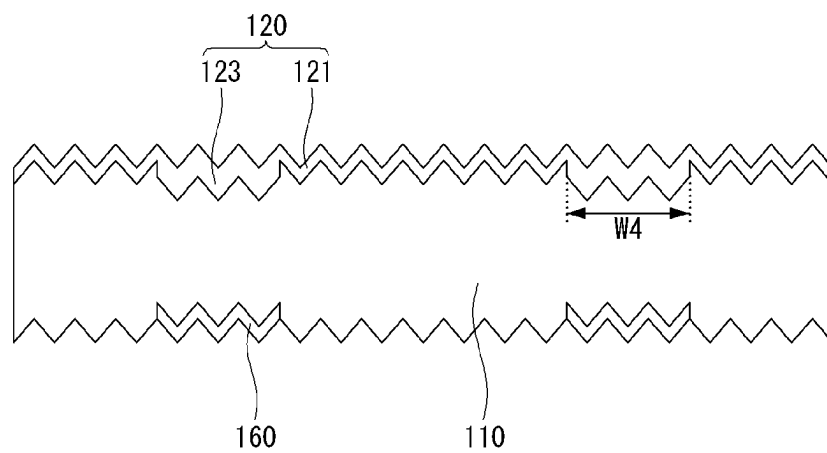


FIG. 5

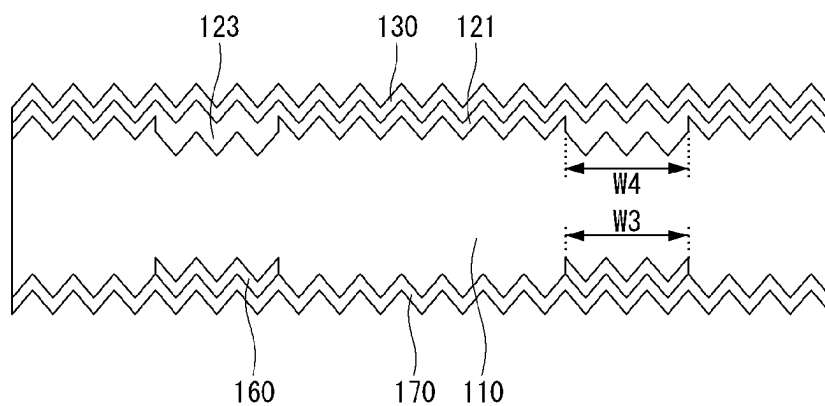


FIG. 6

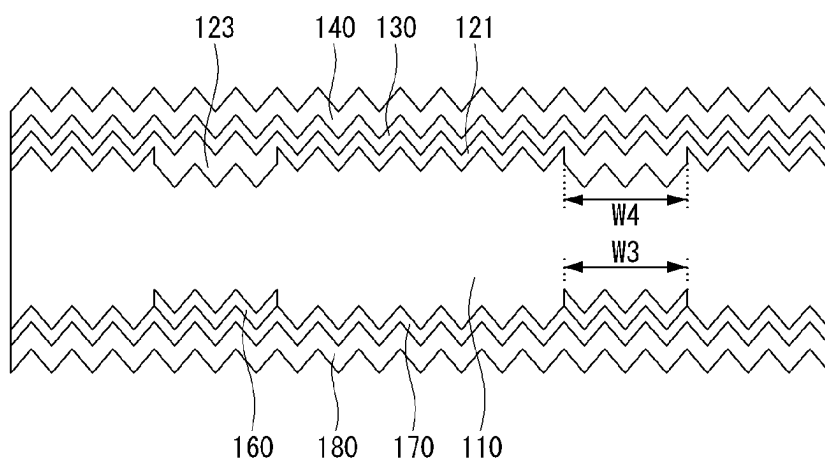
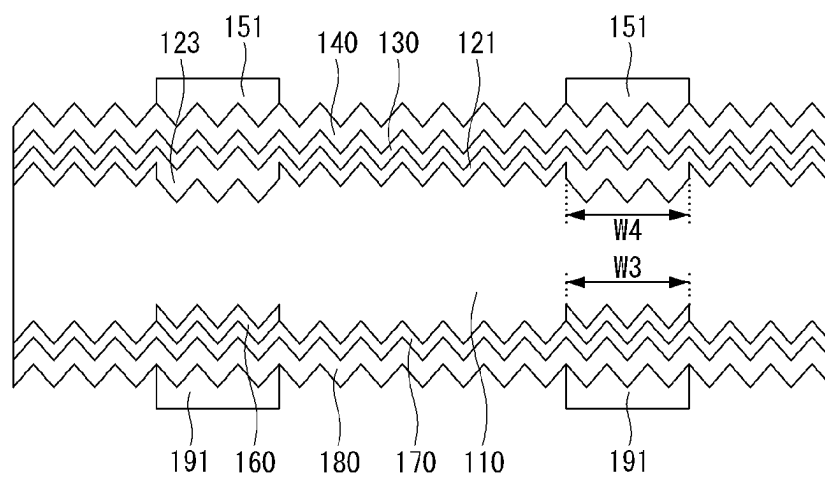


FIG. 7



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**SOLAR CELL****CROSS-REFERENCE TO RELATED APPLICATIONS**

This Application is a Continuation of U.S. patent application Ser. No. 16/560,384, filed on Sep. 4, 2019, which is a Continuation of U.S. patent application Ser. No. 15/443,331, filed on Feb. 27, 2017 (U.S. Pat. No. 10,446,697, issued on Oct. 15, 2019), which is a Continuation of U.S. patent application Ser. No. 13/182,203 filed on Jul. 13, 2011 (U.S. Pat. No. 9,608,133, issued on Mar. 28, 2017), which claims the benefit under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2011-0005429 filed on Jan. 19, 2011, all of these applications are hereby expressly incorporated by reference into the present application.

**BACKGROUND OF THE INVENTION****Field of the Invention**

Embodiments of the invention relate to a solar cell.

**Discussion of the Related Art**

Solar power generation to convert light energy into electric energy using a photoelectric conversion effect has been widely used as a method for obtaining eco-friendly energy. A solar power generation system using a plurality of solar cell panels has been installed in places, such as houses due to an improvement in a photoelectric conversion efficiency of solar cells.

The solar cell generally includes a substrate and an emitter layer that forms a p-n junction along with the substrate, thereby generating an electric current from light incident on the solar cell through one surface of the substrate. Further, an anti-reflection layer is formed on a light receiving surface of the substrate, so as to reduce a reflectance of light incident on the substrate and increase a light transmittance of a predetermined wavelength band. Hence, the photoelectric conversion efficiency of the solar cell increases.

Because light is generally incident on the solar cell through only one surface of the substrate, current conversion efficiency of the solar cell is low. Accordingly, a double-sided light receiving solar cell, in which light is incident on the solar cell through both surfaces of the substrate, has been recently developed.

**SUMMARY OF THE INVENTION**

Embodiments of the invention provide a solar cell with high efficiency.

In one aspect, there is a solar cell including a substrate of a first conductive type, an emitter layer positioned at one surface of the substrate, the emitter layer being a second conductive type opposite the first conductive type, a first electrode electrically connected to the emitter layer, a first protective layer positioned on a front surface of the emitter layer where the first electrode is not positioned, a back surface field layer positioned at another surface of the substrate, a second electrode electrically connected to the back surface field layer, and a second protective layer positioned on a back surface of the substrate where the second electrode is not positioned, wherein each of the first protective layer and the second protective layer is formed of a material having fixed charges of the same conductive type

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as the first conductive type, wherein the back surface field layer is locally positioned at the back surface of the substrate.

Each of the first protective layer and the second protective layer contains aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ) having negative fixed charges. Each of the first protective layer and the second protective layer has a refractive index of about 1.55 to 1.7 and a thickness of about 5 nm to 30 nm.

The back surface field layer is formed at the same location of the substrate as the second electrode and has a surface resistivity of about 30  $\Omega/\text{sq}$  to 80  $\Omega/\text{sq}$ . A width of the back surface field layer may be equal to or less than a width of the second electrode, or may be greater than a width of the second electrode.

The first anti-reflection layer may be formed of silicon nitride ( $\text{SiN}_x$ ) having positive fixed charges.

The solar cell may further include a second anti-reflection layer positioned on a back surface of the second protective layer. The second anti-reflection layer may be formed of silicon nitride ( $\text{SiN}_x$ ) having positive fixed charges.

Each of the first anti-reflection layer and the second anti-reflection layer may have a refractive index of about 1.9 to 2.3 and a thickness of about 50 nm to 100 nm.

The substrate may have a first textured surface corresponding to the surface of the substrate, at which the emitter layer is positioned, and a second textured surface corresponding to the surface of the substrate, at which the back surface field layer is positioned.

The first electrode and the second electrode may be formed of different materials. For example, the first electrode may be formed of a conductive paste containing a mixture (Ag:Al) of silver (Ag) and aluminum (Al), and the second electrode may be formed of a conductive paste containing silver (Ag).

The first electrode and the second electrode may have the same width.

The substrate is formed of phosphorus (P)-doped n-type silicon wafer and has resistivity of about 1  $\Omega\cdot\text{cm}^2$  to 10  $\Omega\cdot\text{cm}^2$ .

The emitter layer may include a first doped region, which is lightly doped with impurities of the second conductive type, and a second doped region, which is more heavily doped than the first doped region with impurities of the second conductive type.

The second doped region may be formed at the same location of the substrate as the first electrode. A width of the second doped region may be equal to or less than a width of the first electrode, or may be greater than a width of the first electrode.

The first doped region has a surface resistivity of about 80  $\Omega/\text{sq}$  to 200  $\Omega/\text{sq}$ , and the second doped region has a surface resistivity of about 30  $\Omega/\text{sq}$  to 80  $\Omega/\text{sq}$ .

According to the above-described characteristics, aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ) for forming the first protective layer has excellent chemical passivation characteristic resulting from a low interface trap density and excellent field effect passivation characteristic resulting from the negative fixed charges. Further, the material is very excellent in stability, moisture transmission, and abrasion resistance. Thus, the first protective layer may reduce a surface recombination velocity, thereby improving the efficiency and the reliability of the solar cell.

Because the back surface field layer is locally positioned at only the same location of the substrate as the second electrode, an influence of the second protective layer formed

of aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ) on carriers moving to the back surface field layer may be minimized.

The second doped region of the emitter layer is formed at only the same location of the substrate as the first electrode, and the first doped region having an impurity concentration lower than the second doped region is formed in the remaining area. Therefore, the solar cell according to the embodiment of the invention may further reduce the recombination of carriers than a solar cell including an emitter layer, whose the entire region is a heavily doped region. Hence, the solar cell according to the embodiment of the invention may secure a low serial resistance.

Both a front surface of the substrate, on which the first electrode is positioned, and a back surface of the substrate, on which the second electrode is positioned, are the textured surfaces, the first protective layer and the first anti-reflection layer are formed on the front surface of the substrate, and the second protective layer and the second anti-reflection layer are formed on the back surface of the substrate. Therefore, light, which is incident on the front surface of the substrate and then passes through the substrate, is again incident on the back surface of the substrate. As a result, the light may be used to generate electric current. Thus, the efficiency of the solar cell according to the embodiment of the invention may further increase than a solar cell generating electric current using only light incident on the front surface of the substrate.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and together with the description serve to explain the principles of the invention. In the drawings:

FIG. 1 is a schematic cross-sectional view of a solar cell according to a first embodiment of the invention;

FIG. 2 is a schematic cross-sectional view of a solar cell according to a second embodiment of the invention; and

FIGS. 3 to 7 are cross-sectional views illustrating a method for manufacturing the solar cell shown in FIG. 2.

#### DETAILED DESCRIPTION OF THE EMBODIMENTS

Embodiments of the invention will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein.

In the drawings, the thickness of layers, films, panels, regions, etc., are exaggerated for clarity. Like reference numerals designate like elements throughout the specification. It will be understood that when an element such as a layer, film, region, or substrate is referred to as being "on" another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present. Further, it will be understood that when an element such as a layer, film, region, or substrate is referred to as being "entirely" on another element, it may be on the entire surface of the other element and may not be on a portion of an edge of the other element.

A solar cell according to an example embodiment of the invention is described in detail with reference to the accompanying drawings.

FIG. 1 is a schematic cross-sectional view of a solar cell according to a first embodiment of the invention.

As shown in FIG. 1, a solar cell according to a first embodiment of the invention includes a substrate 110, an emitter layer 120 positioned at one surface of the substrate 110, for example, a front surface of the substrate 110, a first protective layer 130 positioned on the emitter layer 120, a first anti-reflection layer 140 positioned on the first protective layer 130, a plurality of first electrodes 150 positioned on the emitter layer 120 where the first protective layer 130 and the first anti-reflection layer 140 are not positioned, a back surface field (BSF) layer 160 positioned at a back surface of the substrate 110, a second protective layer 170 positioned on the back surface of the substrate 110, a second anti-reflection layer 180 positioned on a back surface of the second protective layer 170, and a plurality of second electrodes 190 positioned on a back surface of the back surface field layer 160 where the second protective layer 170 and the second anti-reflection layer 180 are not positioned.

The substrate 110 may be formed of a silicon wafer of a first conductive type, for example, n-type, though not required. Silicon used in the substrate 110 may be crystalline silicon, such as single crystal silicon and polycrystalline silicon, or amorphous silicon. When the substrate 110 is of the n-type, the substrate 110 may contain impurities of a group V element such as phosphorus (P), arsenic (As), and antimony (Sb). The substrate 110 may have resistivity of about  $1 \Omega\text{-cm}^2$  to  $10 \Omega\text{-cm}^2$ .

The surface of the substrate 110 may be uniformly textured to form a textured surface corresponding to an uneven surface or having uneven characteristics. More specifically, the substrate 110 has a first textured surface 111 (refer to FIG. 3) corresponding to the front surface, at which the emitter layer 120 is positioned, and a second textured surface 113 (refer to FIG. 3) corresponding to the back surface opposite the front surface.

The emitter layer 120 positioned at the first textured surface 111 of the substrate 110 is an impurity region of a second conductive type (for example, p-type) opposite the first conductive type (for example, n-type) of the substrate 110 and forms a p-n junction along with the substrate 110.

A plurality of electron-hole pairs produced by light incident on the substrate 110 are separated into electrons and holes by a built-in potential difference resulting from the p-n junction between the substrate 110 and the emitter layer 120. The separated electrons move to the n-type semiconductor, and the separated holes move to the p-type semiconductor. When the substrate 110 is of the n-type and the emitter layer 120 is of the p-type, the separated electrons and the separated holes move to the substrate 110 and the emitter layer 120, respectively.

When the emitter layer 120 is of the p-type, the emitter layer 120 may be formed by doping the substrate 110 with impurities of a group III element such as boron (B), gallium (Ga), and indium (In). The emitter layer 120 may have a surface resistivity of about  $30 \Omega/\text{sq}$  to  $120 \Omega/\text{sq}$ .

The first protective layer 130 positioned on the emitter layer 120 formed at the front surface of the substrate 110 is formed of a material having negative fixed charges, for example, aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ). The material of the first protective layer 130 has excellent chemical passivation characteristic resulting from a low interface trap density and excellent field effect passivation characteristic resulting from the negative fixed charges.

Further, the material is very excellent in stability, moisture transmission, and abrasion resistance. Thus, the first protective layer **130** reduces a surface recombination velocity, thereby improving the efficiency and the reliability of the solar cell. In an embodiment of the invention, the first protective layer **130** is formed of a material having fixed charges of the same conductive type as the first conductive type.

The first anti-reflection layer **140** positioned on the first protective layer **130** is formed of a material having positive fixed charges, for example, silicon nitride ( $\text{SiN}_x$ ). The first anti-reflection layer **140** reduces a reflectance of light incident on the solar cell through the front surface of the substrate **110** and increases selectivity of a predetermined wavelength band, thereby increasing the efficiency of the solar cell.

In the embodiment of the invention, the first protective layer **130** has a refractive index of about 1.55 to 1.7 and a thickness  $T_1$  of about 5 nm to 30 nm, and the first anti-reflection layer **140** has a refractive index of about 1.9 to 2.3 and a thickness  $T_2$  of about 50 nm to 100 nm, so as to minimize the light reflectance at the front surface of the substrate **110**.

According to an experiment conducted by the present inventors, the light reflectance at the front surface of the substrate **110** represented a minimum value when the first protective layer **130** and the first anti-reflection layer **140** were within the above refractive index and thickness ranges.

A silicon oxide layer having a thickness of about 1 nm to 3 nm may be further formed at an interface between the first protective layer **130** and the emitter layer **120**.

The plurality of first electrodes **150** are positioned on the emitter layer **120** of the front surface of the substrate **110** and are electrically and physically connected to the emitter layer **120**. The first electrodes **150** extend substantially parallel to one another in a fixed direction. The first electrodes **150** collect carriers (for example, holes) moving to the emitter layer **120**.

The first electrodes **150** may be formed of at least one conductive material selected from the group consisting of nickel (Ni), copper (Cu), tin (Sn), zinc (Zn), indium (In), titanium (Ti), gold (Au), and a combination thereof. In the embodiment of the invention, the first electrodes **150** is formed of a conductive paste obtained by mixing a mixture (Ag:Al) of silver (Ag) and aluminum (Al) with a glass frit.

Alternatively, the first electrodes **150** may be formed using a plating process. In this instance, the first electrode **150** may include a metal seed layer and a conductive layer plated on the metal seed layer. A diffusion prevention layer may be further formed between the metal seed layer and the conductive layer, thereby preventing a material forming the conductive layer from being diffused into the doped region through the metal seed layer.

The metal seed layer may be formed of nickel silicide or aluminum silicide. The conductive layer may contain at least one material selected from the group consisting of copper (Cu), silver (Ag), aluminum (Al), tin (Sn), zinc (Zn), indium (In), titanium (Ti), gold (Au), and a combination thereof.

The plurality of second electrodes **190** are formed on the back surface of the substrate **110** in the same structure as the first electrodes **150**. The second electrodes **190** having the above-described structure collect carriers (for example, electrons) moving to the substrate **110** and output the carriers to an external device.

The second electrodes **190** may be formed of at least one conductive material selected from the group consisting of aluminum (Al), nickel (Ni), copper (Cu), silver (Ag), tin

(Sn), zinc (Zn), indium (In), titanium (Ti), gold (Au), and a combination thereof. In the embodiment of the invention, the second electrodes **190** are formed of a conductive paste obtained by mixing Ag with a glass frit. Alternatively, the second electrodes **190** may be formed using the plating process in the same manner as the first electrodes **150**.

In the embodiment of the invention, a width  $W_1$  of the first electrode **150** is substantially equal to a width  $W_2$  of the second electrode **190** (i.e.,  $W_1=W_2$ ).

The back surface field layer **160** electrically and physically connected to the second electrodes **190** is locally positioned at the back surface of the substrate **110** at the same location as the second electrodes **190**. The back surface field layer **160** is a region (for example,  $n^+$ -type region) that is more heavily doped than the substrate **110** with impurities of the same conductive type as the substrate **110**. In the embodiment of the invention, the back surface field layer **160** has a surface resistivity of about 30  $\Omega/\text{sq}$  to 80  $\Omega/\text{sq}$ . A width  $W_3$  of the back surface field layer **160** may be equal to or less than the width  $W_2$  of the second electrode **190** or may be greater than the width  $W_2$  of the second electrode **190**.

The movement of holes to the back surface of the substrate **110** is prevented or reduced by a potential barrier resulting from a difference between impurity concentrations of the substrate **110** and the back surface field layer **160**. Hence, a recombination and/or a disappearance of electrons and holes around the surface of the substrate **110** are prevented or reduced.

The second protective layer **170** and the second anti-reflection layer **180** are positioned on the back surface of the substrate **110** where the second electrodes **190** and the back surface field layer **160** are not positioned.

In the embodiment of the invention, the second protective layer **170** is formed of the same material as the first protective layer **130** and has the same thickness as the first protective layer **130**. Further, the second anti-reflection layer **180** is formed of the same material as the first anti-reflection layer **140** and has the same thickness as the first anti-reflection layer **140**. Nevertheless, different materials or other materials may be used.

Because the second protective layer **170** positioned on the back surface of the substrate **110** is formed of a material having negative fixed charges, for example, aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ), carriers (for example, electrons) moving to the back surface of the substrate **110** is affected by the second protective layer **170**. However, in the embodiment of the invention, because the back surface field layer **160** is locally positioned at only the same location as the second electrodes **190**, an influence of the second protective layer **170** formed of aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ) on carriers moving to the back surface field layer **160** may be minimized. In an embodiment of the invention, the second protective layer **170** is formed of a material having fixed charges of the same conductive type as the first conductive type.

The solar cell having the above-described structure according to the embodiment of the invention may serve as a double-sided (or bifacial) light receiving solar cell, and an operation of the solar cell is described below.

When light irradiated onto the solar cell is incident on the substrate **110** through the emitter layer **120** and/or the back surface of the substrate **110**, a plurality of electron-hole pairs are generated in the substrate **110** by the light energy. In this instance, because the front surface and the back surface of the substrate **110** are the first textured surface **111** and the second textured surface **113**, respectively, the light reflect-

tance at each of the front surface and the back surface of the substrate **110** may be reduced. Further, because both a light incident operation and a light reflection operation may be performed on each of the first and second textured surfaces **111** and **113** of the substrate **110**, light may be confined in the solar cell. Hence, light absorption may increase, and the efficiency of the solar cell may be improved. In addition, because a reflection loss of the light incident on the substrate **110** may be reduced by the first protective layer **130** and the first anti-reflection layer **140** positioned on the front surface of the substrate **110** and the second protective layer **170** and the second anti-reflection layer **180** positioned on the back surface of the substrate **110**, an amount of light incident on the substrate **110** may further increase.

The electron-hole pairs are separated into electrons and holes by the p-n junction between the substrate **110** and the emitter layer **120**, and the separated holes move to the p-type emitter layer **120** and the separated electrons move to the n-type substrate **110**. The holes moving to the emitter layer **120** move to the first electrodes **150**, and the electrons moving to the substrate **110** move to the second electrodes **190** through the back surface field layer **160**. Accordingly, when the first electrodes **150** of one solar cell are connected to the second electrodes **190** of another solar cell adjacent to the one solar cell using electric wires such as interconnectors, electric current flows through the solar cells and allows use of the electric current for electric power.

The solar cell having the above-described configuration may be used in a state where the solar cell is positioned between a light transmission front substrate and a light transmission back substrate and is sealed by a protective layer.

A solar cell according to a second embodiment of the invention is described below with reference to FIG. 2.

Since the solar cell according to the second embodiment of the invention is substantially the same as the solar cell according to the first embodiment of the invention except an emitter layer, a further description may be briefly made or may be entirely omitted. Hereinafter, only a configuration of the emitter layer is described.

The emitter layer **120** according to the first embodiment of the invention has a uniform doping concentration throughout the entire area of the emitter layer **120**. Thus, the emitter layer **120** according to the first embodiment of the invention may be easily manufactured through a simple process, but a recombination of carriers of the emitter layer **120** may increase because of a high doping concentration. As a result, an improvement in the efficiency of the solar cell may be limited.

Accordingly, the emitter layer **120** according to the second embodiment of the invention is configured so that the recombination of carriers of the emitter layer **120** is prevented or reduced. More specifically, the emitter layer **120** contacting first electrodes **150** is a heavily doped region, and the emitter layer **120** not contacting the first electrodes **150** is a lightly doped region.

In other words, the emitter layer **120** according to the second embodiment of the invention includes a first doped region **121** corresponding to a lightly doped p-type region and a second doped region **123**, which is more heavily doped than the first doped region **121** with p-type impurities.

The second doped region **123** may be formed at the same location of the substrate **110** as the first electrodes **150**. A width  $W_4$  of the second doped region **123** may be equal to or less than a width  $W_1$  of the first electrode **150** or may be greater than the width  $W_1$  of the first electrode **150**.

The first doped region **121** has a surface resistivity of about  $80\ \Omega/\text{sq}$  to  $200\ \Omega/\text{sq}$ , and the second doped region **123** has a surface resistivity of about  $30\ \Omega/\text{sq}$  to  $80\ \Omega/\text{sq}$ .

Accordingly, the solar cell according to the second embodiment of the invention prevents or reduces the recombination of carriers, and thus the efficiency of the solar cell according to the second embodiment of the invention may be further improved.

A method for manufacturing the solar cell according to the second embodiment of the invention is described below with reference to FIGS. 3 to 7.

A substrate **110** formed of a silicon wafer is generally manufactured by slicing a silicon block or an ingot using a blade or a multi-wire saw.

More specifically, the silicon wafer is prepared and then is doped with impurities of a group V element, for example, phosphorus (P) to manufacture the substrate **110** having resistivity of about  $1\ \Omega\cdot\text{cm}^2$  to  $10\ \Omega\cdot\text{cm}^2$ .

When the silicon block or the ingot is sliced, a mechanical damage layer may be formed in the silicon wafer. Thus, a wet etching process for removing the mechanical damage layer is performed, so as to prevent a reduction in characteristics of the solar cell resulting from the mechanical damage layer. An alkaline etchant or an acid etchant may be used in the wet etching process.

After the mechanical damage layer is removed, the wet etching process or a dry plasma etching process is performed to form a first textured surface **111** in the front surface of the substrate **110** and a second textured surface **113** in the back surface of the substrate **110**.

Next, impurities of a group III element are injected into one surface (for example, the first textured surface **111**) of the substrate **110** using an ion implantation method to form an emitter layer **120**.

Because it is easy to control a production amount of ions and an ion velocity moving to the substrate **110** in the ion implantation method, an amount of ions implanted into the substrate **110** and an ion implantation depth when the substrate **110** is doped with impurities using the ion implantation method are controlled more easily than when the substrate **110** is doped with impurities using a thermal diffusion method.

When the emitter **120** is formed using the ion implantation method, the ion implantation depth may vary depending on an amount of ions produced in a process room, an ion implantation energy, and the like.

In this instance, the ion implantation energy may be about 100 KeV to 3 MeV, and the ion implantation depth based on the ion implantation energy of about 100 KeV to 3 MeV may be about  $0.5\ \mu\text{m}$  to  $10\ \mu\text{m}$  measured from the surface of the substrate **110**.

When the emitter layer **120** is formed using the ion implantation method, impurities are lightly injected into the entire front surface of the substrate **110** to form a first doped region **121**.

Next, impurities are heavily injected into a portion (for example, an area to form first electrodes **150**) of the front surface of the substrate **110** to form a second doped region **123**.

In this instance, the first doped region **121** has the surface resistivity of about  $80\ \Omega/\text{sq}$  to  $200\ \Omega/\text{sq}$ , and the second doped region **123** has the surface resistivity of about  $30\ \Omega/\text{sq}$  to  $80\ \Omega/\text{sq}$ .

The emitter layer **120** including the first doped region **121** and the second doped region **123** may be formed using the thermal diffusion method in another embodiment of the invention.

Impurities of a group V element are injected into a portion (for example, an area to form second electrodes **190**) of the other surface (for example, the second textured surface **113**) of the substrate **110** to form a back surface field layer **160** having a surface resistivity of about 30  $\Omega/\text{sq}$  to 80  $\Omega/\text{sq}$ . Further, the ion implantation method may be used to form the back surface field layer **160** in the same manner as the emitter layer **120**.

Subsequently, a natural oxide layer is removed by etching the substrate **110** using hydrofluoric acid (HF). Further, aluminum oxide ( $\text{AlO}_x$ ) or yttrium oxide ( $\text{Y}_2\text{O}_3$ ) is deposited on the front surface and the back surface of the substrate **110** to form a first protective layer **130** and a second protective layer **170**. The first protective layer **130** and the second protective layer **170** may be formed using a plasma enhanced chemical vapor deposition (PECVD) method, a sputtering method, or other methods. In this instance, the first protective layer **130** and the second protective layer **170** are formed so that they have a refractive index of about 1.55 to 1.7 and a thickness of about 5 nm to 30 nm.

When the first protective layer **130** and the second protective layer **170** are formed, a silicon oxide layer having a thickness of about 1 nm to 3 nm may be formed at interfaces between the first protective layer **130** and the substrate **110**, and the second protective layer **170** and the substrate **110**.

After the first protective layer **130** and the second protective layer **170** are formed, silicon nitride ( $\text{SiN}_x$ ) is deposited on a first anti-reflection layer **140** and a second anti-reflection layer **180**. The first anti-reflection layer **140** and the second anti-reflection layer **180** may be formed using the PECVD method, the sputtering method, or other methods.

In this instance, the first anti-reflection layer **140** and the second anti-reflection layer **180** are formed so that they have a refractive index of about 1.9 to 2.3 and a thickness of about 50 nm to 100 nm. The second anti-reflection layer **180** formed on the back surface of the substrate **100** may be thicker than the first anti-reflection layer **140** formed on the front surface of the substrate **100**.

Subsequently, a first conductive paste **151** obtained by mixing a mixture (Ag:Al) of silver (Ag) and aluminum (Al) with a glass frit is printed on the front surface of the substrate **100** with a first electrode pattern, and a second conductive paste **191** obtained by mixing Ag with the glass frit is printed on the back surface of the substrate **100** with a second electrode pattern. A firing process is then performed on the first conductive paste **151** and the second conductive paste **191**.

In the firing process, the material of the protective layers **130** and **170** and the material of the anti-reflection layers **140** and **180** are etched by an etching component contained in the glass frit. Hence, the first electrodes **150** electrically and physically connected to the second doped region **123** of the emitter layer **120** are formed, and the second electrodes **190** electrically and physically connected to the back surface field layer **160** are formed.

Alternatively, the first electrodes **150** and the second electrodes **190** may be formed using a plating process.

A portion of the surface of the substrate **110** to form the first electrodes **150** and the second electrodes **190** has to be exposed so as to form the first electrodes **150** and the second electrodes **190** using the plating process. Thus, the same portion between the first protective layer **130** and the first anti-reflection layer **140** and the same portion between the second protective layer **170** and the second anti-reflection layer **180** may be removed using a dry etching process to form the first electrodes **150** and the second electrodes **190** on the removed portions of the surface of the substrate **110**.

When the first electrodes **150** and the second electrodes **190** are formed, a metal seed layer may be formed on the exposed portions of the surface of the substrate **110** and then a conductive layer or a diffusion prevention layer and a conductive layer may be formed on the metal seed layer.

In embodiments of the invention, reference to fixed charges includes oxide fixed charges.

Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

What is claimed is:

1. A bifacial solar cell comprising:

an n-type substrate, the n-type substrate having a front surface and a back surface;

a p-type emitter layer positioned at the front surface of the n-type substrate and forming a p-n junction along with the n-type substrate;

a first protective layer positioned on the emitter layer;

a first anti-reflection layer positioned on the first protective layer;

a second protective layer positioned on the back surface of the n-type substrate;

a second anti-reflection layer positioned on a back surface of the second protective layer,

a first silicon oxide layer positioned between the first protective layer and the emitter layer, and

a second silicon oxide layer positioned between the second protective layer and the back surface of the n-type substrate,

wherein the first protective layer and the second protective layer are formed of aluminum oxide.

2. The bifacial solar cell of claim 1, further comprising:

a plurality of back surface field layers partially positioned at the back surface of the n-type substrate.

3. The bifacial solar cell of claim 2, wherein each of the plurality of back surface field layers is an n-type region that is more heavily doped than the n-type substrate.

4. The bifacial solar cell of claim 2, wherein the second protective layer and the plurality of back surface field layers are not overlapping each other.

5. The bifacial solar cell of claim 2, further comprising:

a plurality of first electrodes partially positioned on the p-type emitter layer and electrically connected to the p-type emitter layer; and

a plurality of second electrodes positioned on the plurality of back surface field layers and electrically connected to the plurality of back surface field layers.

6. The bifacial solar cell of claim 5, wherein a width of a back surface field layer of the plurality of back surface field layers is equal to or less than a width of a second electrode of the plurality of second electrodes.

7. The bifacial solar cell of claim 5, wherein lower surfaces of the plurality of first electrodes entirely and directly contact the p-type emitter layer.

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8. The bifacial solar cell of claim 5, wherein the plurality of first electrodes are formed of different compositions from the plurality of second electrodes.
9. The bifacial solar cell of claim 5, wherein the plurality of first electrodes are formed of a first conductive paste including a glass frit and a mixture (Ag:Al) of silver (Ag) and aluminum (Al), and the plurality of second electrodes are formed of a second conductive paste including a glass frit and silver (Ag), and
- wherein each of the plurality of second electrodes protrudes through a hole in the second protective layer.
10. The bifacial solar cell of claim 5, wherein the first protective layer and the second protective layer have a same thickness.
11. The bifacial solar cell of claim 1, wherein the first silicon oxide layer has a thickness of about 1 nm to 3 nm; and/or the second silicon oxide layer has a thickness of about 1 nm to 3 nm.
12. The bifacial solar cell of claim 5, wherein the plurality of first electrodes are parallel to each other and collect carriers moving to the p-type emitter layer, and the plurality of second electrodes are parallel to each other and collect carriers moving to the n-type substrate.
13. The bifacial solar cell of claim 1, wherein the p-type emitter layer includes a first doped region, which is doped with impurities of the p-type, and a second doped region, which is more heavily doped than the first doped region with impurities of the p-type.
14. The bifacial solar cell of claim 13, wherein the second doped region is formed at the same location of the substrate as the plurality of first electrodes.

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15. The bifacial solar cell of claim 13, wherein the first doped region has a surface resistivity of about 80  $\Omega/\text{sq}$  to 200  $\Omega/\text{sq}$ , and the second doped region has a surface resistivity of about 30  $\Omega/\text{sq}$  to 80  $\Omega/\text{sq}$ .
16. The bifacial solar cell of claim 1, wherein the first anti-reflection layer contains silicon nitride.
17. The bifacial solar cell of claim 1, wherein the first protective layer contains negative fixed charges, and the first anti-reflection layer and the second anti-reflection layer contain positive fixed charges.
18. The bifacial solar cell of claim 1, wherein the first protective layer has a refractive index of about 1.55 to 1.7 and a thickness of about 5 nm to 30 nm, and the first anti-reflection layer and the second anti-reflection layer have a refractive index of about 1.9 to 2.3 and a thickness of about 50 nm to 100 nm.
19. A bifacial solar cell comprising:  
 an n-type substrate, the n-type substrate having a front surface and a back surface;  
 a p-type emitter layer positioned at the front surface of the n-type substrate and forming a p-n junction along with the n-type substrate;  
 a first protective layer positioned on the emitter layer;  
 a first anti-reflection layer positioned on the first protective layer;  
 a second protective layer positioned on the back surface of the n-type substrate;  
 a second anti-reflection layer positioned on a back surface of the second protective layer,  
 a first silicon oxide layer positioned between the first protective layer and the emitter layer, and  
 a second silicon oxide layer positioned between the second protective layer and the back surface of the n-type substrate,  
 wherein the second protective layer is formed of aluminum oxide.

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